

Advanced Plane Geometry



Circle theorems: tangent-chord and secant angles

- 1 Two secants are drawn from an external point to a circle, intercepting arcs of 100° and 30° . Find the measure of the angle formed at the external point, using the formula $(\text{difference of arcs}) \div 2$.
 - 2 Two chords intersect inside a circle, intercepting arcs of 80° and 40° . Find the measure of the angle formed at the intersection, using the formula $(\text{sum of arcs}) \div 2$.
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Power of a point (secant and tangent segment relationships)

- 3 A tangent segment from an external point has length 8, and a secant from the same point passes through the circle with external segment length 4. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.
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Area of regular polygons

- 4 Find the area of a regular pentagon with side length 6 and apothem 4.13, using $A = \frac{1}{2} \times \text{perimeter} \times \text{apothem}$.
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Similar solids and scale factor in 3D

- 5 Two similar cylinders have a scale factor of $\frac{2}{3}$. If the smaller cylinder has volume 40, find the volume of the larger cylinder.
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Composite 3D solids

- 6 A solid consists of a cylinder of radius 3 and height 8, topped with a hemisphere of the same radius. Find the total volume, in terms of π .
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Triangle centers: centroid and circumcenter

- 7 Find the centroid of the triangle with vertices (2, 3), (8, 3), (5, 9), using the formula (average of the vertex coordinates).
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Heron's formula

- 8 A triangle has sides 7, 8, and 9. Find its area using Heron's formula, $A = \sqrt{s(s-a)(s-b)(s-c)}$ where s is the semi-perimeter.
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A multi-part circle geometry problem

- 9 This question has two parts. A circle has radius 10. A chord is drawn 6 units from the center. (a) Find the length of the chord, using the relationship between the radius, the distance from center to chord, and half the chord length (a right triangle relationship). (b) Find the area of the minor segment cut off by this chord, to the nearest whole number, given that the central angle subtended by the chord is approximately 106.3° and $\sin(106.3^\circ) \approx 0.960$.
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Angle bisectors and the angle bisector theorem

- 10 In triangle ABC , the angle bisector from A meets side BC at point D , dividing it such that $BD = 6$ and $DC = 9$. If $AB = 8$, find AC , using the angle bisector theorem ($\frac{BD}{DC} = \frac{AB}{AC}$).
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The 30-60-90 and 45-45-90 triangles in composite figures

- 11 An equilateral triangle has side length 10. Find its height and area, using the fact that the height splits it into two 30-60-90 triangles.
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Circumscribed and inscribed circles

- 12 A square is inscribed in a circle of radius 5. Find the side length of the square, using the fact that the diagonal of the square equals the diameter of the circle.
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Regular polygon interior and exterior angles

- 13 Find the measure of each interior angle and each exterior angle of a regular octagon.
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A multi-part composite figure and circle problem

- 14** This question has three parts. A running track consists of a rectangle with two semicircular ends. The rectangle is 80 meters long and 40 meters wide, and the semicircles are attached to the two 40-meter-wide ends. (a) Find the radius of each semicircle. (b) Find the total perimeter (distance around) the track, in terms of π . (c) Find the total enclosed area of the track shape, in terms of π .